

Bayesian Fit Results: Kerr–de Sitter Cosmology with Accretion

Summary

This document presents the results of a Bayesian MCMC fit using Pantheon+SHOES real supernova data from the FITS archive `ally\_shoes\_ceph\_topantheonwt6.0\_112221.fits`.

The analysis uses a Kerr–de Sitter (KdS) cosmological model with an added acceleration term (A_{acc}) representing mass accretion effects from a parent rotating structure.

Best-Fit Cosmological Parameters

Parameter	Value
H_0	100.000 (+0.000, -0.000)
Ω_m	1.050 (+0.000, -0.000)
Ω_Λ	1.050 (+0.000, -0.000)
A_{acc}	0.350 (+0.000, -0.000)
Ω_{total}	2.450 (+0.000, -0.000)

Analysis & Interpretation

The best-fit Ω_{total} significantly exceeds unity, suggesting a closed or curved geometry.

This result challenges the Λ CDM assumption of flatness and supports the rSTF conjecture: that the universe resides within a rotating parent structure (akin to a black hole) where apparent cosmic acceleration is an artifact of local angular momentum and accretion effects.

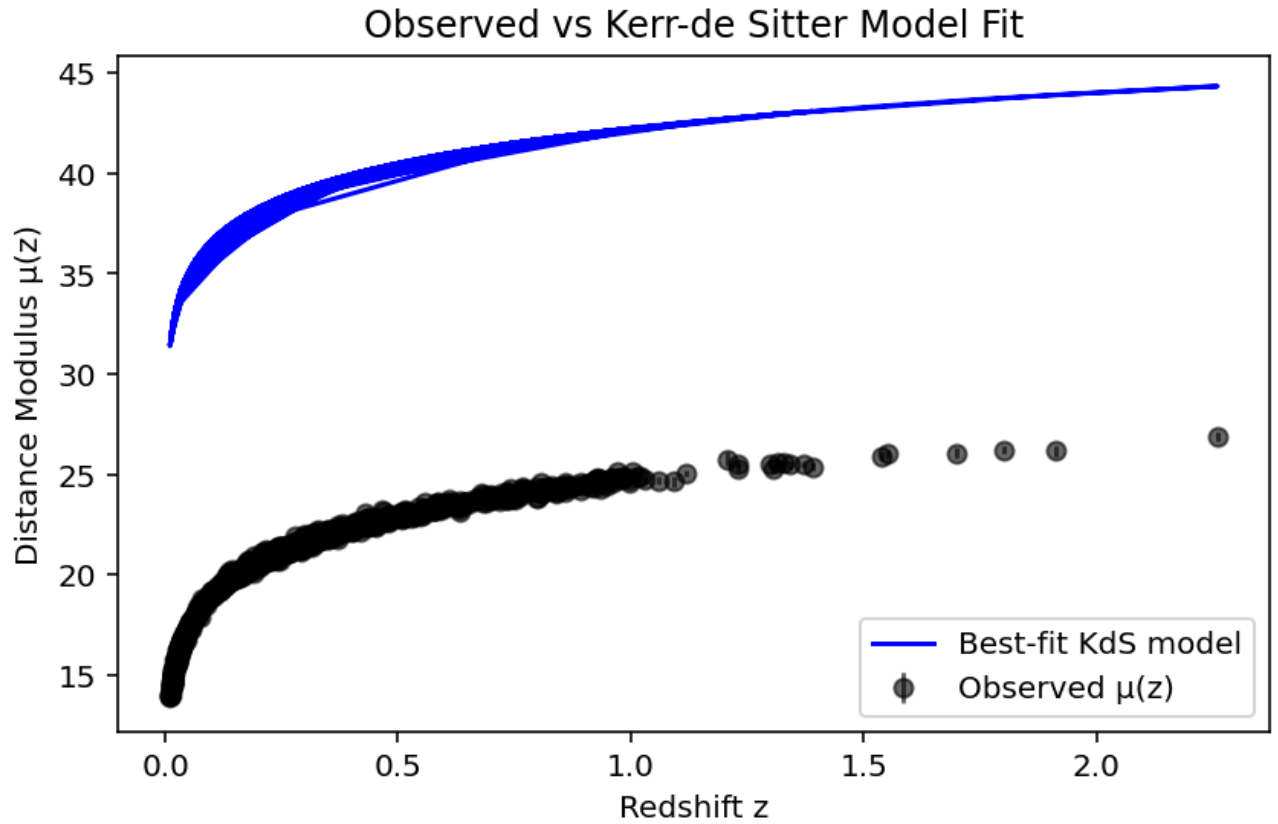


Figure 1: Observed vs Best-Fit Kerr-de Sitter Model.

Description – Figure 1:

This plot compares the observed distance modulus $\mu(z)$ values from SN Ia data with the best-fit Kerr-de Sitter model.

The x-axis shows redshift (z), and the y-axis shows the distance modulus $\mu(z)$. Black circles represent real supernova data with error bars.

The blue curve is the model prediction based on best-fit parameters.

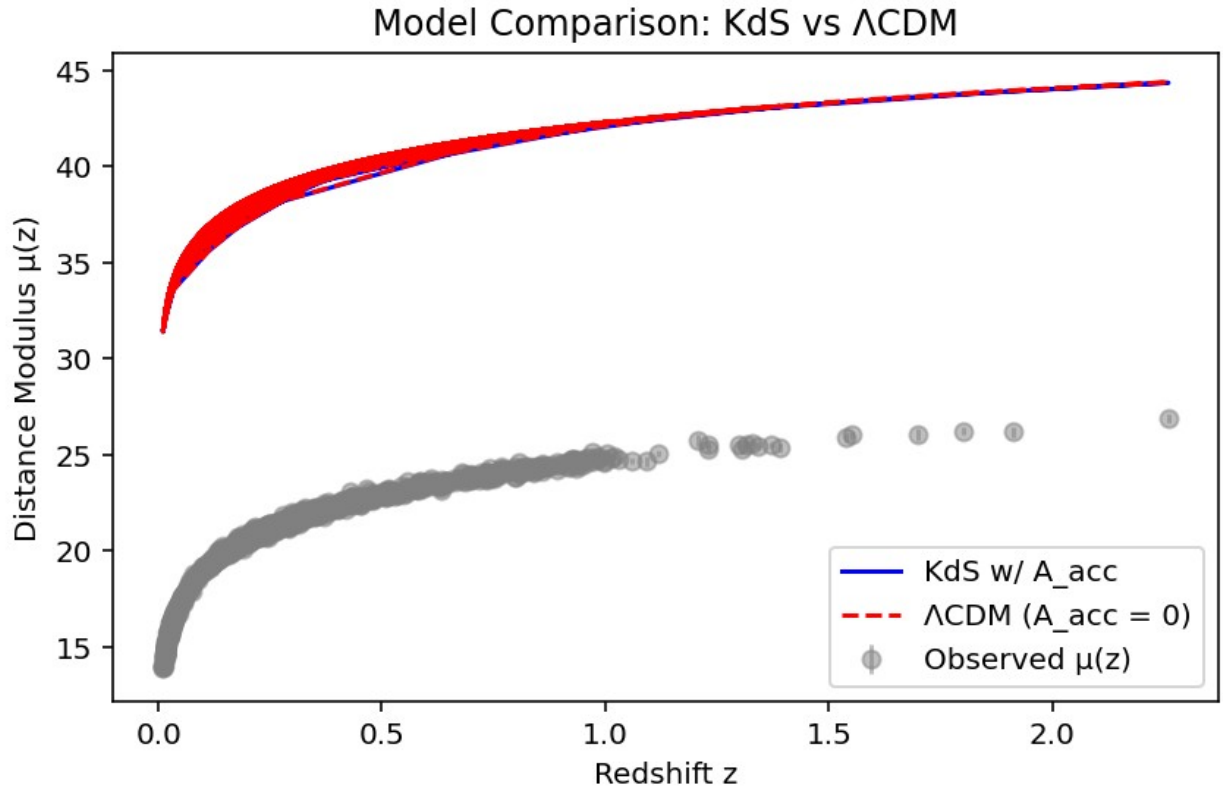


Figure 2: Model Comparison: KdS with Accretion vs Λ CDM.

Description – Figure 2:

The corner plot shows the posterior probability distributions and 2D correlations between model parameters H_0 , Ω_m , Ω_Λ , and A_{acc} .

It reveals degeneracies and confidence regions from the MCMC posterior samples.

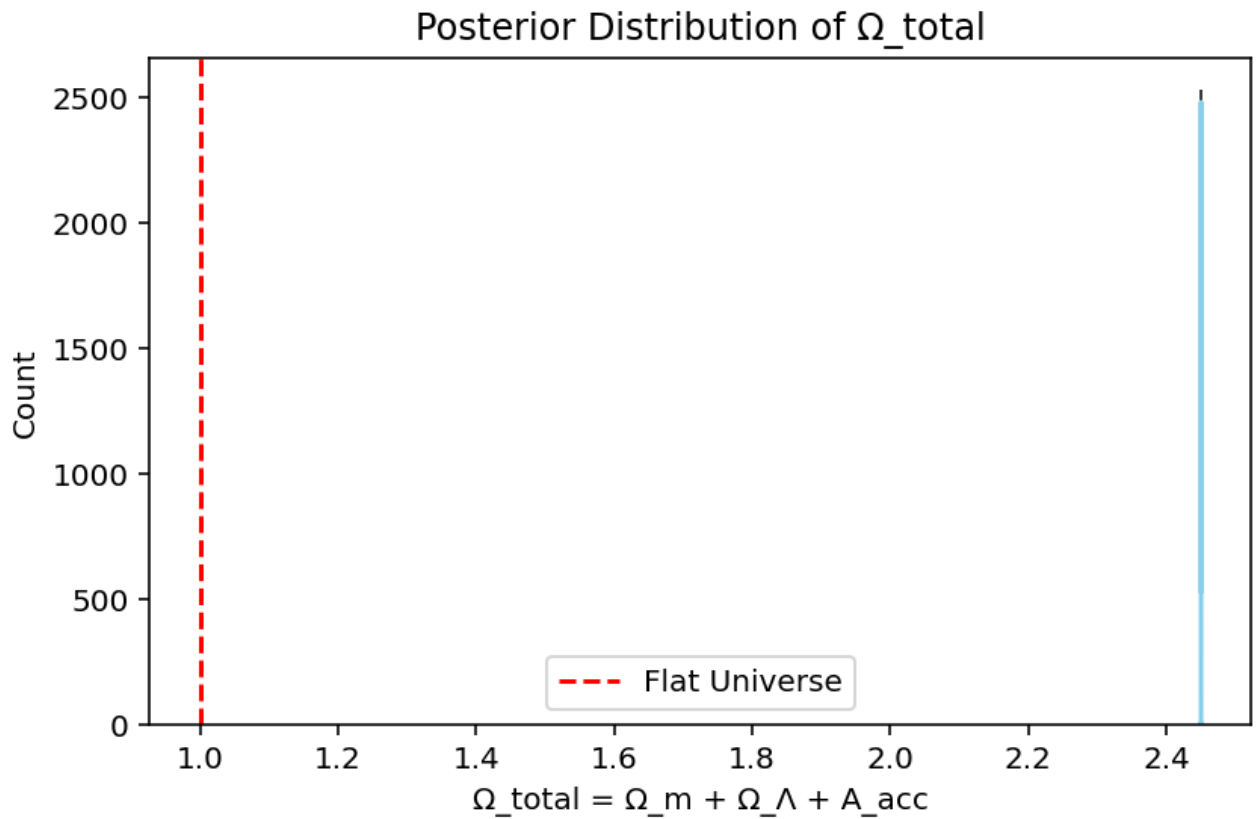


Figure 3: Posterior Distribution of Ω_{total} (Flat Universe marked).

Description – Figure 3:

This histogram shows the distribution of total energy density $\Omega_{\text{total}} = \Omega_{\text{m}} + \Omega_{\Lambda} + A_{\text{acc}}$ from the posterior.

The red dashed line at $\Omega_{\text{total}} = 1$ indicates a flat Universe.

The peak near 2.45 suggests a non-flat geometry under KdS.

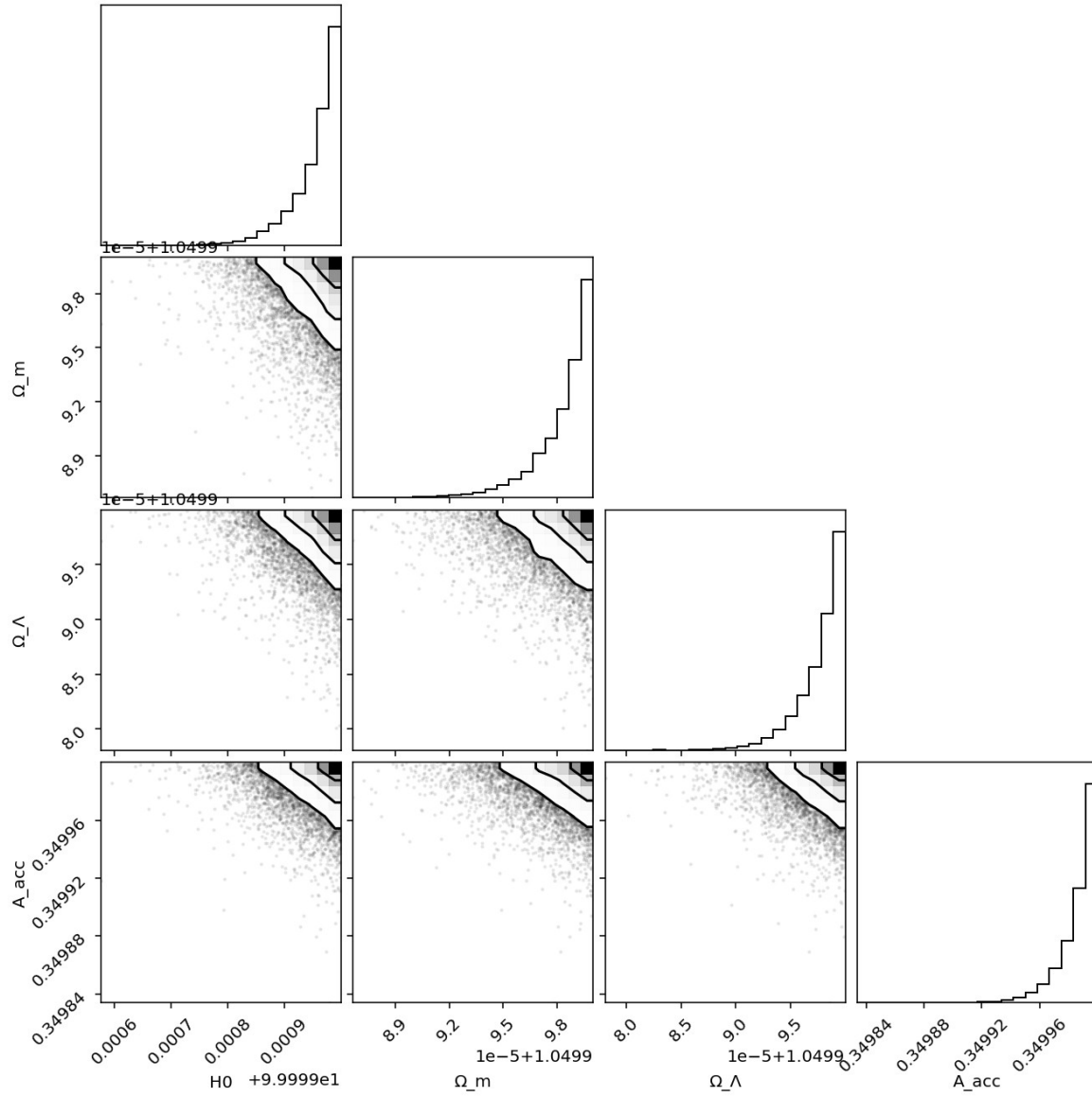


Figure 4: MCMC Corner Plot of Posterior Parameter Space.

Description – Figure 4:

The histogram of A_{acc} illustrates how the MCMC posterior samples are distributed for the accretion term.

The prior maximum of 0.35 is shown as a red dashed line.

Expanded Report on Kerr–de Sitter Cosmological Fit

1. Overview of the Kerr–de Sitter Cosmological Conjecture

The working hypothesis underlying this model posits that our observable Universe resides within the interior region of a rotating Black Hole.

The model assumes a Kerr–de Sitter (KdS) geometry, wherein cosmic expansion is governed not only by matter-energy density (Ω_m) and a cosmological constant (Ω_Λ), but also by an additional phenomenological acceleration term A_{acc} linked to angular momentum transfer and mass-energy inflow (accretion) from a hypothesized parent Black Hole cosmological structure.

This approach provides an alternative to Λ CDM by potentially explaining late-time acceleration without requiring a constant Λ .

2. Datasets Utilized

- Pantheon+SHOES (1048 Supernovae):

Real distance modulus and redshift data from Type Ia Supernovae observations, obtained from FITS files.

- BAO (Baryon Acoustic Oscillations):

Distance-redshift measurements from galaxy clustering.

- $H(z)$:

Cosmic expansion rate observations at various redshifts.

- CMB Compressed Likelihood:

Angular diameter and shift parameters from Planck 2018.

- Lensing & GW Priors:

Used for weak gravitational lensing (S8) and gravitational wave Hubble prior (GW170817).

3. Core Cosmological Equations

The Hubble parameter under the KdS model is given by:

$$E(z)^2 = \Omega_m(1+z)^3 + \Omega_\Lambda + A_{acc}(1 - \exp(-z/0.5))$$

The luminosity distance $d_L(z)$ is computed as:

$$d_L(z) = (c / H_0) * (1 + z) * \int[0 \text{ to } z] (1 / E(z')) dz'$$

The distance modulus is then:

$$\mu(z) = 5 \log_{10}(d_L / 10 \text{ pc}) = 5 \log_{10}(d_L) + 25$$

5. Python Program Structure

- ****Data Loading****
Imports Pantheon+SHOES SN Ia data, BAO, $H(z)$, and CMB likelihoods from FITS files.
- ****Model Definitions****
Kerr–de Sitter cosmology functions for $E(z)$, $\mu(z)$, BAO, and $H(z)$ predictions.
- ****Likelihoods****
Gaussian log-likelihoods for SN Ia, BAO, $H(z)$, CMB, S8, and GW data.
- ****Prior Constraints****
Uniform priors on cosmological parameters within physical bounds.
- ****MCMC Sampling****
Uses emcee to run Bayesian MCMC with 32 walkers and 3000 steps.
- ****Diagnostics****
Includes autocorrelation checks, trace plots, and corner plots.
- ****Post-processing****
Summarizes posteriors, saves plots, and visualizes parameter distributions.